Arian Salari

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Education

University of California, Irvine — **Irvine, CA**
B.S. in Mechanical Engineering, Honors Collegium — *September 2026 – Expected June 2028*

Irvine Valley College — **Irvine, CA**
Associate's in Engineering, Honors Program, Dean's List, Cumulative GPA: 3.95 — *August 2024 – May 2026*

Engineering Experience

Rezvani Motors — **Dec 2024 – Mar 2025**
Mechanical Engineering Intern — *Irvine, CA*

- Updated SolidWorks CAD models and technical drawings for vehicle components, incorporating physical measurements and calculations to support design revisions and prototype development.
- Performed CFD-based aerodynamic evaluations to assess airflow behavior and support vehicle design decisions.
- Supported prototype development and manufacturing by reviewing drawings, inspecting components for fitment, and communicating design observations with engineering and production teams.

Projects & Research

Autonomous Solar-Assisted Quadcopter UAV, Personal Project — **May 2026 – Present**

- Designed a parametric quadcopter airframe in SolidWorks and 3D-printed custom structural components for fitment testing, assembly, and iterative design refinement.
- Built a solar-powered autonomous charging station for a ∼74 Wh battery system, enabling self-recharge between flights without manual battery replacement.
- Integrating ∼20 W of lightweight onboard photovoltaics with autonomous return-to-dock functionality to reduce battery power demand and extend operational autonomy.

Aircraft Winglet Aerodynamics and Efficiency, IVC / Saddleback Research Symposium — **Aug 2025 – Jan 2026**

- Designed and 3D-printed baseline, sharklet, and split-scimitar wingtip configurations for comparative aerodynamic testing in physical and digital wind-tunnel environments.
- Quantified relative wingtip-vortex strength using wake-string deflection and AI-assisted image analysis, with the split-scimitar reducing average deflection from 1.53 in to 0.98 in, a ∼36% reduction versus the baseline.
- Extended prior award-winning research that earned Second Place in Fluids and Aerodynamics at the Orange County Science and Engineering Fair.

Medical Device Validation Research, IVC / Saddleback Research Symposium — **Aug 2024 – Jan 2025**

- Investigated Normothermic Ex-Vivo Heart Perfusion (NEHP) as a physiological testing platform for artificial heart valve validation under controlled flow and pressure conditions.
- Compared FDA-approved mechanical validation methods with industry testing practices, identifying limitations in replicating physiological conditions and evaluating NEHP-perfused porcine hearts as a higher-fidelity testing approach.

Shape-Memory Alloy Robotic Hand, OC Science and Engineering Fair — **Nov 2023 – Mar 2024**

- Designed and built a five-finger robotic hand actuated by Nitinol shape-memory alloy wire, using electrical current to induce contraction and drive finger articulation.
- Refined the actuation mechanism across multiple prototypes, using elastic return elements to restore finger position and improve coordinated motion across all five fingers.

Leadership

Irvine Valley College STEM Club — **Mar 2025 – May 2026**
President — *Irvine, CA*

- Led hands-on engineering projects and competition teams, including rocket-design builds, while organizing technical workshops, industry speaker events, and research/internship opportunities.

Irvine Valley College Computer Science Club — **May 2025 – May 2026**
Vice President — *Irvine, CA*

- Organized programming workshops, technical speaker sessions, and hackathon preparation while connecting members with internships and research opportunities.

Technical Skills

CAD & Design: SolidWorks, AutoCAD, Fusion 360, Mechanical Design, Technical Drawings, DFM/DFA
Analysis & Simulation: CFD, SolidWorks Flow Simulation, MATLAB, Data Analysis
Programming: Python, C/C++, JavaScript
Manufacturing & Prototyping: 3D Printing, Rapid Prototyping, Fitment Testing, Design Iteration
Tools: Excel, GitHub, Visual Studio, Technical Documentation